

Final Assessment

Scenario 1

You are working as a Supply Chain Officer within the United States Centers for Disease Control and Prevention (US CDC). You are provided with the health commodity shipment data for July.

You have been asked the following questions by the head of supply chain. Use the data analytics concepts you have learned in the course to answer them.

1. What is/are the main issue(s)/concern(s) for the July health commodity shipments?
2. What is the potential cause(s) of the identified concern(s)?
3. What can you expect for the month of August if no decision is made?
4. What actions would you recommend to address the identified concern(s)? In your recommendations, consider the following
 - a. Mode of transportation
 - b. Natural disaster assessment
 - c. Punctuality and timeliness
 - d. Road and traffic

To answer the questions:

- Use the data provided in the worksheet “Raw data” of the workbook “US CDC Shipment.xlsx”.
- You are also provided with a data dictionary that describes each variable.
- For each question:
 - Provide clear written explanation
 - Use tables, charts, plots, and numbers to justify your answer
 - Your problem identification, root cause identification, predictions for August and recommendations should be based on insights extracted from the data (combine with Supply Chain knowledge and experience, if needed).
- Your answer should be submitted in a report format (in Word or Excel) with tables and charts that can be submitted to the US CDC Head of Supply Chain.

Good Luck!!

Scenario 2

You are the Supply Chain Planner at **Med-Supply.org**¹. You are provided with the consolidated delivery data from 2006 to 2015, and you are asked to perform the following tasks. Use the data in the sheet “Raw data” of the “**Med-Supply-org.xlsx**” workbook to provide your answer.

1. Data preparation and cleaning

Some data are not captured properly, including dates combined with text, values combined with text and so on. Before starting the analysis, your data should be cleaned, with columns containing one type of data at once. This means, if the column is related to date, it should not have text. The same applies to the column related to price, cost, or amount.

- Use your analytical judgement to identify the column where data has been wrongly captured and use your creativity to provide an adapted solution without losing any information.
- Please provide details on which columns you have identified with wrong data entry, what was/were the exact issue(s) and how you addressed it/them.
- Feel free to format the cells and data type, and perform any other data preparation tasks you noticed might be necessary. However, you have to provide details on each task you have performed to keep track of the changes.

2. Data Analysis and insights extraction

a. Details on the supported projects

- i. What was the total number of health projects that Med-Supply.org was supporting?
- ii. In how many countries were these projects being implemented?
- iii. What was the country with the highest number of projects?
- iv. Which hub was managing the highest number of countries and how many countries was it?
- v. Aside from the hub that was managing the highest number of countries, which countries (name) were the other hubs managing?
- vi. List the countries that were fulfilled via Direct drop and the ones that were fulfilled via Regional Distribution Center (RDC).
- vii. What was the primary shipment mode used when for projects managed by field Offices, and what were the top three countries to which shipments were delivered during the period of analysis?

¹ **Med-Supply.org** is a fictive International Organization with four hubs (PMO – US, South Africa Field Office, Haiti Field Office, Ethiopia Field Office) providing Health Supply Chain service all over the world. It has been created for the sake of this case study.

b. Service efficiency

By analyzing the delivery-related dates:

- i. How long did the shipments take (on average of working days) to be delivered by truck?
- ii. For a particular shipment, could we expect that it would be delivered before the scheduled date, on time, or with a lot of delays after the scheduled delivery date?
- iii. Based on the insights above, what can we conclude about the efficiency of the delivery services?
- iv. Was the delivery recording process as efficient as the delivery process itself? Explain why.

c. Details on the products

- i. Based on the sub classification of the products, how many countries **Med-Supply.org** has supported in terms of response to Malaria?
- ii. What was the name of the main vendor used for HIV test and what were the top four brands used for this category of products?
- iii. What was the total weight of all the products shipped to Latin America (Belize, Guatemala, Guyana, Haiti)?

3. Data visualization

a. Mapping

Represent on a Map:

- i. The total number of projects supported by country.
- ii. The total Freight Cost (USD) per supported country

b. Heatmap

- i. Create a Heatmap, highlighting the number of projects that each hub was supporting in each country.

Good luck!